

教學實作：擴充 OLED 溫濕度感測器功能

AMB82-mini ： SDA: PIN 1, SCL: PIN 2
HUB 8735 Ultra ： SDA: PIN 1, SCL: PIN 2
ESP32-S3 ： SDA: PIN 1, SCL: PIN 2

步驟 1：修改硬體定義提示詞 (Devices Definition)

讓 AI 認識 Oled 感測器及其物理特性。

- Text Display (OLED Display SSD1306, I2C):
 - SDA: GPIO 1
 - SCL: GPIO 2
 - Resolution: 128x64 pixels
 - Display lines: 4
 - Supports UTF-8 text (including Traditional Chinese, English, numbers, and symbols)
 - Maximum 21 ASCII characters per line (actual capacity depends on character width and font)
 - Writable only
 - Supports text display only

步驟 2：定義工具 JSON 規範 (Tools Definition)

在 toolsDefinition 中加入 /dht11 指令的格式規範。

OLED display control:

```
{
  "type": "tool_call",
  "method": "/oled",
  "params": {
    "line1": "<Text for line 1. Empty string clears the line.>",
    "line2": "<Text for line 2. Empty string clears the line.>",
    "line3": "<Text for line 3. Empty string clears the line.>",
    "line4": "<Text for line 4. Empty string clears the line.>"
  }
}
```

Success response:

```
{
  "status": "success",
  "method": "/oled",
  "workId": "<system-provided>"
}
```

Error response:

```
{
  "status": "error",
  "method": "/oled",
  "reason": "<error reason>",
  "workId": "<system-provided>"
}
```

步驟 3：底層 C++ 程式碼實作

在專案中新增 `tool_dht11` 驅動函數。

```
#include <U8g2lib.h>    // U8g2 2.27.6
#include <Wire.h>

const int OLED_SDA = 1;
const int OLED_SCL = 2;
U8G2_SSD1306_128X64_NONAME_F_SW_I2C u8g2(U8G2_R0, /* clock=*/ OLED_SCL, /* data=*/ OLED_SDA, /* reset=*/
U8X8_PIN_NONE);

String tool_oled(String line1, String line2, String line3, String line4, String workId) {

    u8g2.clear();
    u8g2.clearBuffer();
    u8g2.setFont(u8g2_font_unifont_t_chinese1);

    if (line1 != "")
        u8g2.drawUTF8(0, 16, line1.c_str());
    if (line2 != "")
        u8g2.drawUTF8(0, 32, line2.c_str());
    if (line3 != "")
        u8g2.drawUTF8(0, 48, line3.c_str());
    if (line4 != "")
        u8g2.drawUTF8(0, 64, line4.c_str());

    u8g2.sendBuffer();

    return
        "{\"status\": \"success\", \""
        "\"method\": \"/oled\", \""
        "\"workId\": \"" + workId + "\"}";
}

void setup() {
    u8g2.begin();
    u8g2.setFont(u8g2_font_10x20_me);
    u8g2.enableUTF8Print();
    u8g2.clear();
}
```

步驟 4：更新工具分發器邏輯 (executeTool)

將 /dht11 方法註冊進系統的分發流程。

```
void executeTool(String workId, String command, JsonObject params, bool reCheck = true) {

    if (command == "/digitalwrite" || command == "/analogwrite") {
        // ... 原有內容不變 ...

    }

    else if (command == "/oled") {

        String line1 = params["line1"] | "";
        String line2 = params["line2"] | "";
        String line3 = params["line3"] | "";
        String line4 = params["line4"] | "";

        String response = tool_oled(line1, line2, line3, line4, workId );

        if (xSemaphoreTake(stateMutex, MUTEX_TIMEOUT_TICKS) == pdTRUE) {

            historicalMessages += buildGeminiMessage("user", command + timestamps);
            historicalMessages += buildGeminiMessage("model", response + timestamps);

            executeToolHistory += workId + " " + command + " [ " + line1 + " | " + line2 + " | " + line3 + " | " + line4 + " ]\n";

            xSemaphoreGive(stateMutex);
        }

        evaluateWorkflowContinuation(workId, reCheck);
    }

    else if (command == "/digitalread" || command == "/analogread") {
        // ... 原有內容不變 ...
    }
    // ... 其餘 tools 不變 ...
}
```